

Problem: Universal Functional Divisibility

Let a class of functions $\mathcal{F}$ be defined as follows. Every function $f \in \mathcal{F}$ satisfies

$$f : \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{>0}$$

and

$$f(m) + n \mid m^2 + f(m)f(n)$$

for all positive integers m, n .

Determine the minimum possible value of $f(100)$.